

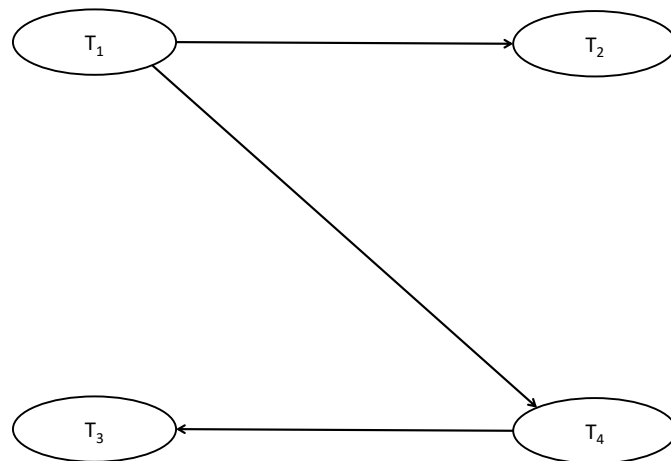
1. Consider the concurrent transactions T_1 , T_2 , T_3 and T_4 under the schedule S in Figure 2, which access relational tables $Student(stuID, stuName, age, height)$ and $Grade(stuID, cosID, grade)$ concurrently. The tuples/rows in the tables are viewed as the data items to be locked by DBMS and independently accessed by the transactions.

- (1) Construct the precedence graph for S .
- (2) Is S a serializable schedule? If not, give the reason. If it is, give a serial schedule that is equivalent to S .
- (3) Is S a recoverable schedule, and why?
- (4) Is S a cascadeless schedule, and why?

T_1	T_2	T_3	T_4
begin_transaction			
	begin_transaction		
update <i>Student</i> set <i>stuName</i> ='Li' where <i>stuID</i> =10			
		begin_transaction	
	update <i>Student</i> set <i>age</i> = <i>age</i> +1 where <i>stuID</i> =20		
		select <i>age</i> from <i>Student</i> where <i>stuID</i> =50	
			begin_transaction
			update <i>Student</i> set <i>height</i> = <i>height</i> +5 where <i>stuID</i> =30
update <i>Grade</i> set <i>grade</i> = <i>grade</i> +1 where <i>stuID</i> =40 AND <i>CosID</i> =10			
	select <i>stuName</i> from <i>Student</i> where <i>stuID</i> =10		
	commit		
commit			
			select <i>stuID, grade</i> from <i>Grade</i> where <i>stuID</i> =40 AND <i>CosID</i> =10
			commit
		Select <i>height</i> from <i>Student</i> where <i>stuID</i> =30	
		commit	

Figure 2 Schedule S

(1) 前驱图



说明：

T₁ 修改了 student 表中 *stuID*=10 的元组, T₂ 随后又读取该元组, 因此存在一条从到的冲突边;
对其它各条冲突边可做类似分析。

(2) 前驱图中无回路, 因此是冲突可串行的。

(3) 不是可恢复调度。

T₂ 读取 student 表中 *stuID*=10 的元组时, 该元组内容已由 T₁ 修改过, 但 T₂ 提交操作 **commit 早于 T₁ 提交 commit**。一旦 T₁ 在 T₂ 的 commit 操作之后回滚其 update student 操作, 将 *stuName* 回滚为旧值, 则 T₂ 的 select 操作无法随着回滚, T₂ 读取的仍然是 *stuName* 修改后的值 (1 分)。

(4) 不是无级联回滚调度

因为不是可恢复调度, 因此也不是无级联回滚调度。

或者: 无级联回滚调度中, 一个事务只能读取其它事务已经提交的数据。

但 T₂ 读取 student 表中 *stuID*=10 的元组时, 该元组被被 T₁ 修改, 但未提交。

2. The schedule S in Fig. 3 controls the transaction T_1 , T_2 , T_3 and T_4 to concurrently access the database *University*.

T_1	T_2	T_3	T_4
begin trans			
update <i>Instructor</i> set <i>salary</i> =2200 where <i>ID</i> =2019			
	begin trans		
	update <i>Student</i> set <i>age</i> = <i>age</i> +2 where <i>sid</i> =211		
checkpoint			
commit			
		begin trans	
		update <i>Course</i> set <i>credit</i> =3 where <i>title</i> =DBS	
			begin trans
			update <i>Department</i> set <i>budget</i> =20000 where <i>dname</i> =CompSci
checkpoint			
	rollback		
			update <i>Instructor</i> set <i>salary</i> = <i>salary</i> +200 where <i>ID</i> =2019
			commit
		select <i>credit</i> from <i>Course</i> where <i>ID</i> =211	
system crash			

Figure 3 Schedule S

These transactions modify the following data items:

- 1) the value of the attribute *salary* of the *instructor* whose *ID* is 2019 in *Instructor*(*ID*, *name*, *department*, *salary*),
- 2) the value of the attribute *age* of the *student* whose *s_id* is 211 in *Student*(*sid*, *name*, *department*, *age*),
- 3) the value of the attribute *credit* of the course *Database System Concepts* (i.e. DBS) in *Course*(*course_id*, *title*, *department*, *credit*), and
- 4) the value of the attribute *budget* of the department *Computer Science* (i.e. CompSci) in

Department(dname, building, budget).

It is assumed that

- i) the initial values of the attribute *salary*, *age*, *credit* and *budget* to be updated are 2000, 18, 0, and 10000, respectively; and
- ii) immediate database modification and checkpoint techniques are employed.

Q&A:

(1) Give the log records of the schedule *S*.

<T1, start>

<T1, salary, 2000, 2200>

<T2, start>

<T2, age, 18, 20>

Checkpoint{T1, T2}

<T1, commit>

<T3, start>

<T3, credit, 0, 3>

<T4, start>

<T4, budget, 10000, 20000>

Checkpoint{T2,T3,T4}

<T2, age, 18> /rollback 产生的日志

<T2, abort> /rollback 产生的日志

<T4, salary, 2200, 2400>

<T4, commit>

****crash****

(2) After the second checkpoint, what are the values of the data items *salary*, *age*, *credit* and *budget* in the database?

salary=2200, age=20, credit=3, and budget=20000

(3) After the system crash occurs, what recovery actions (i.e. redo, undo, ignore) should be conducted respectively for T₁, T₂, T₃, and T₄?

T₁: ignore

T₂: ignore or redo

T₃: undo,

T₄: redo

(4) After recovery operations on T₁, T₂, T₃, and T₄ are completed, what are the values of the data items *salary*, *age*, *credit* and *budget* in the database?

salary=2400, age=18, credit=0, and budget=20000